

# Rotational Motion of a Rigid Reference Frame

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This document describes the computations that are performed by the class `RigidReferenceFrame` and its subclasses to determine the rotational motion (rotation, angular velocity, and angular acceleration) of a rigid frame.

## Definitions

We consider in this document a rigid reference frame defined by two bodies  $B_1$  and  $B_2$  at positions  $\mathbf{q}_1$  and  $\mathbf{q}_2$ , respectively. We construct a basis of the reference frame using three vectors having the following properties:

- the *fore* vector  $\mathbf{F}$  which is along the axis  $\mathbf{q}_2 - \mathbf{q}_1$ ;
- the *normal* vector  $\mathbf{N}$  which is orthogonal to  $\mathbf{F}$  and is such that the velocity  $\dot{\mathbf{q}}_2 - \dot{\mathbf{q}}_1$  is in the plane  $(\mathbf{F}, \mathbf{N})$ ;
- the *binormal* vector  $\mathbf{B}$  which is orthogonal to  $\mathbf{F}$  and  $\mathbf{N}$  such that  $(\mathbf{F}, \mathbf{N}, \mathbf{B})$  forms a direct trihedron.

There are obviously many possible choices for  $(\mathbf{F}, \mathbf{N}, \mathbf{B})$ . In practice, it is convenient to choose  $\mathbf{B}$  before  $\mathbf{N}$  so that the basis is computed exclusively using vector products:

$$\begin{cases} \mathbf{F} &= \mathbf{r} \\ \mathbf{B} &= \mathbf{r} \wedge \ddot{\mathbf{r}} \\ \mathbf{N} &= \mathbf{B} \wedge \mathbf{F} \end{cases} \quad (1)$$

where we have defined  $\mathbf{r} := \mathbf{q}_2 - \mathbf{q}_1$ . Since we'll need to use  $\ddot{\mathbf{r}}$  later, it is important to note here that  $\ddot{\mathbf{r}} = \ddot{\mathbf{q}}_2 - \ddot{\mathbf{q}}_1$  where  $\ddot{\mathbf{q}}_1$  is the acceleration exerted on  $B_1$  by the rest of the system (and similarly,  $\ddot{\mathbf{q}}_2$  is the acceleration exerted on  $B_2$  by the rest of the system).

It is trivial to check that these definitions satisfy the properties above, and in particular that they determine a direct orthogonal basis. The corresponding orthonormal basis is:

$$\begin{cases} \mathbf{f} &= \frac{\mathbf{F}}{|\mathbf{F}|} \\ \mathbf{b} &= \frac{\mathbf{B}}{|\mathbf{B}|} \\ \mathbf{n} &= \frac{\mathbf{N}}{|\mathbf{N}|} \end{cases} \quad (2)$$

These vectors are sufficient to define the rotation of the reference frame at any point in time.

## Derivatives of normalized vectors

In what follows, we will need to compute the time derivatives of the elements of the trihedron  $(\mathbf{f}, \mathbf{n}, \mathbf{b})$ . To help with this we prove two formulæ that define the first and second derivatives of  $\mathbf{V}/|\mathbf{V}|$  based on that of  $\mathbf{V}$ .

The first derivative is:

$$\begin{aligned}
 \frac{d}{dt} \frac{\mathbf{v}}{|\mathbf{v}|} &= \frac{|\mathbf{v}| \dot{\mathbf{v}} - \frac{d|\mathbf{v}|}{dt} \mathbf{v}}{|\mathbf{v}|^2} \\
 &= \frac{|\mathbf{v}| \dot{\mathbf{v}} - \frac{(\mathbf{v} \cdot \dot{\mathbf{v}})}{|\mathbf{v}|} \mathbf{v}}{|\mathbf{v}|^2} \\
 &= \frac{\dot{\mathbf{v}}}{|\mathbf{v}|} - \mathbf{v} \frac{(\mathbf{v} \cdot \dot{\mathbf{v}})}{|\mathbf{v}|^3}
 \end{aligned} \tag{3}$$

The second derivative is somewhat more complicated:

$$\begin{aligned}
 \frac{d^2}{dt^2} \frac{\mathbf{v}}{|\mathbf{v}|} &= \frac{d}{dt} \left( \frac{|\mathbf{v}|^2 \dot{\mathbf{v}} - (\mathbf{v} \cdot \dot{\mathbf{v}}) \mathbf{v}}{|\mathbf{v}|^3} \right) \\
 &= \frac{|\mathbf{v}|^3 \frac{d}{dt} (|\mathbf{v}|^2 \dot{\mathbf{v}} - (\mathbf{v} \cdot \dot{\mathbf{v}}) \mathbf{v}) - 3|\mathbf{v}| (\mathbf{v} \cdot \dot{\mathbf{v}}) (|\mathbf{v}|^2 \dot{\mathbf{v}} - (\mathbf{v} \cdot \dot{\mathbf{v}}) \mathbf{v})}{|\mathbf{v}|^6} \\
 &= \frac{2(\mathbf{v} \cdot \dot{\mathbf{v}}) \dot{\mathbf{v}} + |\mathbf{v}|^2 \ddot{\mathbf{v}} - (|\dot{\mathbf{v}}|^2 + (\mathbf{v} \cdot \ddot{\mathbf{v}})) \mathbf{v} - (\mathbf{v} \cdot \dot{\mathbf{v}}) \dot{\mathbf{v}}}{|\mathbf{v}|^3} - 3 \frac{|\mathbf{v}|^3 (\mathbf{v} \cdot \dot{\mathbf{v}}) \dot{\mathbf{v}} - |\mathbf{v}| (\mathbf{v} \cdot \dot{\mathbf{v}})^2 \mathbf{v}}{|\mathbf{v}|^6} \\
 &= \frac{\ddot{\mathbf{v}}}{|\mathbf{v}|} - 2 \dot{\mathbf{v}} \frac{(\mathbf{v} \cdot \dot{\mathbf{v}})}{|\mathbf{v}|^3} - \mathbf{v} \frac{|\dot{\mathbf{v}}|^2 + (\mathbf{v} \cdot \ddot{\mathbf{v}})}{|\mathbf{v}|^3} + 3 \mathbf{v} \frac{(\mathbf{v} \cdot \dot{\mathbf{v}})^2}{|\mathbf{v}|^5}
 \end{aligned} \tag{4}$$

## Angular velocity

To compute the angular velocity, we start by deriving the vectors (1) and obtain:

$$\begin{cases} \dot{\mathbf{F}} &= \dot{\mathbf{r}} \\ \dot{\mathbf{B}} &= \mathbf{r} \wedge \ddot{\mathbf{r}} \\ \dot{\mathbf{N}} &= \dot{\mathbf{B}} \wedge \mathbf{F} + \mathbf{B} \wedge \dot{\mathbf{F}} \end{cases} \tag{5}$$

Injecting these expressions in the derivative formula (3) makes it possible to compute the trihedron of the derivatives  $(\dot{\mathbf{f}}, \dot{\mathbf{n}}, \dot{\mathbf{b}})$  of (2). The angular velocity  $\boldsymbol{\omega}$  is, by definition, such that:

$$\begin{cases} \boldsymbol{\omega} \wedge \mathbf{f} &= \dot{\mathbf{f}} \\ \boldsymbol{\omega} \wedge \mathbf{n} &= \dot{\mathbf{n}} \\ \boldsymbol{\omega} \wedge \mathbf{b} &= \dot{\mathbf{b}} \end{cases}$$

In the orthonormal basis  $(\mathbf{f}, \mathbf{n}, \mathbf{b})$  let's write  $\boldsymbol{\omega} = \alpha \mathbf{f} + \beta \mathbf{n} + \gamma \mathbf{b}$ . Substituting in the equations above we obtain:

$$\begin{cases} \dot{\mathbf{f}} &= \beta \mathbf{n} \wedge \mathbf{f} + \gamma \mathbf{b} \wedge \mathbf{f} \\ \dot{\mathbf{n}} &= \alpha \mathbf{f} \wedge \mathbf{n} + \gamma \mathbf{b} \wedge \mathbf{n} \\ \dot{\mathbf{b}} &= \alpha \mathbf{f} \wedge \mathbf{b} + \beta \mathbf{n} \wedge \mathbf{b} \end{cases}$$

Now multiply (using the scalar product) the first equation by  $\mathbf{n}$ , the second by  $\mathbf{b}$  and the third by  $\mathbf{f}$ , and use the properties of the triple product to simplify the result using  $(\mathbf{n} \wedge \mathbf{f}) \cdot \mathbf{n} = 0$  and  $(\mathbf{b} \wedge \mathbf{f}) \cdot \mathbf{n} = 1$  (and similar expressions obtained by circular permutation). We find:

$$\begin{cases} \dot{\mathbf{f}} \cdot \mathbf{n} &= \gamma \\ \dot{\mathbf{n}} \cdot \mathbf{b} &= \alpha \\ \dot{\mathbf{b}} \cdot \mathbf{f} &= \beta \end{cases}$$

which finally yields the following expression for the angular velocity:

$$\boldsymbol{\omega} = (\dot{\mathbf{n}} \cdot \mathbf{b}) \mathbf{f} + (\dot{\mathbf{b}} \cdot \mathbf{f}) \mathbf{n} + (\dot{\mathbf{f}} \cdot \mathbf{n}) \mathbf{b}$$

## Angular acceleration

To compute the angular acceleration, we start by deriving the vectors (5) and obtain:

$$\begin{cases} \ddot{\mathbf{r}} &= \ddot{\mathbf{r}} \\ \ddot{\mathbf{B}} &= \dot{\mathbf{r}} \wedge \ddot{\mathbf{r}} + \mathbf{r} \wedge \mathbf{r}^{(3)} \\ \ddot{\mathbf{N}} &= \ddot{\mathbf{B}} \wedge \mathbf{F} + 2\dot{\mathbf{B}} \wedge \dot{\mathbf{F}} + \mathbf{B} \wedge \ddot{\mathbf{F}} \end{cases} \quad (6)$$

Injecting these expressions in the second derivative formula (4) makes it possible to compute the trihedron of the second derivatives  $(\ddot{\mathbf{f}}, \ddot{\mathbf{n}}, \ddot{\mathbf{b}})$  of (2). The angular acceleration is then written:

$$\begin{aligned} \dot{\boldsymbol{\omega}} &= (\ddot{\mathbf{n}} \cdot \mathbf{b})\mathbf{f} + (\dot{\mathbf{n}} \cdot \dot{\mathbf{b}})\mathbf{f} + (\dot{\mathbf{n}} \cdot \mathbf{b})\dot{\mathbf{f}} \\ &+ (\ddot{\mathbf{b}} \cdot \mathbf{f})\mathbf{n} + (\dot{\mathbf{b}} \cdot \dot{\mathbf{f}})\mathbf{n} + (\dot{\mathbf{b}} \cdot \mathbf{f})\dot{\mathbf{n}} \\ &+ (\ddot{\mathbf{f}} \cdot \mathbf{n})\mathbf{b} + (\dot{\mathbf{f}} \cdot \dot{\mathbf{n}})\mathbf{b} + (\dot{\mathbf{f}} \cdot \mathbf{n})\dot{\mathbf{b}} \end{aligned}$$

## Jerk

The alert reader will have noticed the presence of the jerk,  $\mathbf{r}^{(3)}$  in (6). This section explains how it is calculated.

Consider a system of  $n$  massive bodies  $B_k, k \in [1, n]$  located at positions  $\mathbf{q}_k(t)$  at time  $t$ . The acceleration field at point  $\mathbf{q}$  is:

$$\mathbf{r}(\mathbf{q}, \mathbf{q}_k(t)) = \sum_{k=1}^n \frac{\mu_k(\mathbf{q}_k - \mathbf{q})}{|\mathbf{q}_k - \mathbf{q}|^3}$$

The jerk vector field is the total derivative of this field with respect to time:

$$\frac{d\mathbf{r}}{dt} = \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \frac{d\mathbf{q}}{dt} + \sum_{k=1}^n \frac{\partial \mathbf{r}}{\partial \mathbf{q}_k} \frac{d\mathbf{q}_k}{dt} = \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \dot{\mathbf{q}} + \sum_{k=1}^n \frac{\partial \mathbf{r}}{\partial \mathbf{q}_k} \dot{\mathbf{q}}_k$$

To compute it, let's first focus on the partial derivatives with respect to  $\mathbf{q}_k$ :

$$\begin{aligned} \frac{\partial \mathbf{r}}{\partial \mathbf{q}_k} &= \frac{\partial}{\partial \mathbf{q}_k} \frac{\mu_k(\mathbf{q}_k - \mathbf{q})}{|\mathbf{q}_k - \mathbf{q}|^3} \\ &= \mu_k \left( \frac{\frac{\partial}{\partial \mathbf{q}_k}(\mathbf{q}_k - \mathbf{q})}{|\mathbf{q}_k - \mathbf{q}|^3} - 3 \frac{(\mathbf{q}_k - \mathbf{q}) \otimes \frac{\partial}{\partial \mathbf{q}_k} |\mathbf{q}_k - \mathbf{q}|}{|\mathbf{q}_k - \mathbf{q}|^4} \right) \\ &= \mu_k \left( \frac{\mathbb{1}}{|\mathbf{q}_k - \mathbf{q}|^3} - 3 \frac{(\mathbf{q}_k - \mathbf{q}) \otimes \frac{\mathbb{1} \cdot (\mathbf{q}_k - \mathbf{q})}{|\mathbf{q}_k - \mathbf{q}|}}{|\mathbf{q}_k - \mathbf{q}|^4} \right) \\ &= \mu_k \left( \frac{\mathbb{1}}{|\mathbf{q}_k - \mathbf{q}|^3} - 3 \frac{(\mathbf{q}_k - \mathbf{q}) \otimes (\mathbf{q}_k - \mathbf{q})}{|\mathbf{q}_k - \mathbf{q}|^5} \right) \end{aligned}$$

Note that this expression is a symmetric bilinear form, as expected since the gravitational acceleration derives from a potential, and therefore its Jacobian (the Hessian of the potential) is symmetric.

The partial derivative with respect to  $\mathbf{q}$  is then easy to compute by noting that each term in  $\frac{\partial \mathbf{r}}{\partial \mathbf{q}}$  has the same form as the partial derivative above, with a sign change due to deriving with respect to  $\mathbf{q}$  instead of  $\mathbf{q}_k$ :

$$\frac{\partial \mathbf{r}}{\partial \mathbf{q}} = \sum_{k=1}^n \mu_k \left( -\frac{\mathbb{1}}{|\mathbf{q}_k - \mathbf{q}|^3} + 3 \frac{(\mathbf{q}_k - \mathbf{q}) \otimes (\mathbf{q}_k - \mathbf{q})}{|\mathbf{q}_k - \mathbf{q}|^5} \right)$$

Putting all these calculations together we obtain the jerk exercised by the bodies  $B_k$  on point  $\mathbf{q}$  at time  $t$ :

$$\frac{d\mathbf{r}}{dt} = \sum_{k=1}^n \mu_k \left( \frac{\mathbb{1}}{|\mathbf{q}_k - \mathbf{q}|^3} - 3 \frac{(\mathbf{q}_k - \mathbf{q}) \otimes (\mathbf{q}_k - \mathbf{q})}{|\mathbf{q}_k - \mathbf{q}|^5} \right) (\dot{\mathbf{q}}_k - \dot{\mathbf{q}}) \quad (7)$$

The computation of  $\ddot{\mathbf{r}} = \ddot{\mathbf{q}}_2 - \ddot{\mathbf{q}}_1$  is then straightforward.  $\ddot{\mathbf{q}}_1$  is obtained by evaluating (7) at  $\mathbf{q} = \mathbf{q}_1$  using a system from which  $B_1$  is excluded. Similarly,  $\ddot{\mathbf{q}}_2$  is obtained by evaluation (7) at  $\mathbf{q} = \mathbf{q}_2$  using a system from which  $B_2$  is excluded.

## Application to the body surface reference frame

While the formulæ above were derived assuming a rigid reference frame defined by two bodies  $B_1$  and  $B_2$ , they remain valid for a reference frame defined by a single body  $B_1$  provided that we give a proper definition of the vector  $\mathbf{r}$ . We now consider how they apply to the *body surface* reference frame.

It is convenient to choose  $\mathbf{r}$  to be a unit vector rotating with the surface of the body; without loss of generality,  $\mathbf{r}$  may be written:

$$\mathbf{r} = \mathbf{x} \cos(\omega t + \varphi) + \mathbf{y} \sin(\omega t + \varphi)$$

where  $\mathbf{x}$  and  $\mathbf{y}$  form a direct orthonormal basis of the equatorial plane of the body. The time derivatives of  $\mathbf{r}$  are:

$$\begin{cases} \dot{\mathbf{r}} &= -\mathbf{x}\omega \sin(\omega t + \varphi) + \mathbf{y}\omega \cos(\omega t + \varphi) \\ \ddot{\mathbf{r}} &= -\mathbf{x}\omega^2 \cos(\omega t + \varphi) - \mathbf{y}\omega^2 \sin(\omega t + \varphi) = -\omega^2 \mathbf{r} \\ \mathbf{r}^{(3)} &= \mathbf{x}\omega^3 \sin(\omega t + \varphi) - \mathbf{y}\omega^3 \cos(\omega t + \varphi) = -\omega^2 \dot{\mathbf{r}} \end{cases}$$

Using (1) it's easy to see that:

$$\begin{cases} \mathbf{F} &= \mathbf{r} \\ \mathbf{B} &= \omega \mathbf{z} \\ \mathbf{N} &= \dot{\mathbf{r}} \end{cases}$$

where  $\mathbf{z}$  is along the rotation axis such that  $(\mathbf{x}, \mathbf{y}, \mathbf{z})$  form a direct trihedron. The orthonormal vectors follow immediately, noting that  $|\mathbf{r}| = 1$ :

$$\begin{cases} \mathbf{f} &= \mathbf{r} \\ \mathbf{b} &= \mathbf{z} \\ \mathbf{n} &= \frac{\dot{\mathbf{r}}}{\omega} \end{cases}$$

The first derivatives of the orthonormal trihedron can be computed directly:

$$\begin{cases} \dot{\mathbf{f}} &= \dot{\mathbf{r}} \\ \dot{\mathbf{b}} &= 0 \\ \dot{\mathbf{n}} &= \frac{\ddot{\mathbf{r}}}{\omega} = -\omega \mathbf{r} \end{cases}$$

which yields, not surprisingly,  $\boldsymbol{\omega} = \omega \mathbf{z}$ . The second derivatives are equally straightforward:

$$\begin{cases} \ddot{\mathbf{f}} &= \ddot{\mathbf{r}} = -\omega^2 \mathbf{r} \\ \ddot{\mathbf{b}} &= 0 \\ \ddot{\mathbf{n}} &= -\omega \dot{\mathbf{r}} \end{cases}$$

and yield, as expected,  $\dot{\boldsymbol{\omega}} = 0$ .